1) $\begin{cases} 3x+5y+z=-2 \\ -2x-2y-3z=7 \\ x+4y+z=-5 \end{cases}$.

$$\Delta = \begin{vmatrix} 3 & 5 & 1 \\ -2 & -2 & -3 \\ 1 & 4 & 1 \end{vmatrix} = 3(-2)1 + 1\cdot 5\cdot(-3) + 1(-2)4 - 1(-2)1 - 1(-2)5 - 3(-3)4 =$$

$$= -6 - 15 - 8 + 2 + 10 + 36 = 19$$

$\Delta \neq 0$, => единственное решение

$$\Delta x = \begin{vmatrix} -2 & 5 & 1 \\ 7 & -2 & -3 \\ -5 & 4 & 1 \end{vmatrix} = (-2)(-2)1 + 1\cdot 7\cdot 4 + (-3)5(-3) -$$

$$- 1(-2)(-5) - 1\cdot 5\cdot 7 - (-2)(-3)4 =$$

$$= 4 + 28 + 75 - 10 - 35 - 24 =$$

$$= 38$$

$$\Delta y = \begin{vmatrix} 3 & -2 & 1 \\ -2 & 7 & -3 \\ 1 & -5 & 1 \end{vmatrix} = 3\cdot 7\cdot 1 + 1(-2)(-5) + 1(-2)(-3) -$$

$$- 1\cdot 7\cdot 1 - 1(-2)(-2) - 3(-3)(-5) =$$

$$= 21 + 10 + 6 - 7 - 4 - 45 =$$

$$= -19$$

$$\Delta_z = \begin{vmatrix} 3 & 5 & -2 \\ -2 & -2 & 7 \\ 1 & 4 & -5 \end{vmatrix} = 3(-2)(-5) + 1 \cdot 5 \cdot 7 + (-2)(-2)4 -$$

$$-(-2)(-2)1 - (-5)(-2)5 - 3 \cdot 4 \cdot 7 =$$

$$= 30 + 35 + 16 - 4 - 50 - 84 =$$

$$= -57$$

$$x = \frac{\Delta_x}{\Delta} = \frac{38}{19} = 2 \qquad y = \frac{\Delta_y}{\Delta} = \frac{-19}{19} = -1$$

$$z = \frac{\Delta_z}{\Delta} = \frac{-57}{19} = -3$$

Odbem: $(2; -1; -3)$

Проверка

$$\begin{cases} 3x + 5y + z = -2 \\ -2x - 2y - 3z = 7 \\ x + 4y + z = -5 \end{cases}$$

$$\begin{cases} 3 \cdot 2 + 5(-1) + (-3) = -2 \\ -2(2) - 2(-1) - 3(-3) = 7 \\ 2 + 4(-1) + (-3) = -5 \end{cases}$$

$$\begin{cases} 6 - 5 - 3 = -2 \\ -4 + 2 + 9 = 7 \\ 2 - 4 - 3 = -5 \end{cases}$$

2) $\begin{cases} x+y+z=6 \\ 3x-y+3z=2 \\ 4x+3y-z=5 \end{cases}$

$$\Delta = \begin{vmatrix} 1 & 1 & 1 \\ 3 & -1 & 3 \\ 4 & 3 & -1 \end{vmatrix} = 1(-1)(-1) + 1\cdot 3\cdot 3 + 4\cdot 1\cdot 3 - $$

$$-1(-1)4 - (-1)3\cdot 1 - 1\cdot 3\cdot 3 =$$

$$= 1 + 9 + 12 + 4 + 3 - 9 =$$

$$= 20$$

$\Delta \neq 0 \Rightarrow$ единственное решение

$$\Delta x = \begin{vmatrix} 6 & 1 & 1 \\ 2 & -1 & 3 \\ 5 & 3 & -1 \end{vmatrix} = 6(-1)(-1) + 1\cdot 2\cdot 3 + 5\cdot 3\cdot 1 -$$

$$- 1(-1)5 - (-1)2\cdot 1 - 6\cdot 3\cdot 3 =$$

$$= 6 + 6 + 15 + 5 + 2 - 54 =$$

$$= -20$$

$$\Delta y = \begin{vmatrix} 1 & 6 & 1 \\ 3 & 2 & 3 \\ 4 & 5 & -1 \end{vmatrix} = 1\cdot 2\cdot(-1) + 1\cdot 3\cdot 5 + 4\cdot 6\cdot 3 -$$

$$- 1\cdot 2\cdot 4 - (-1)\cdot 3\cdot 6 - 1\cdot 5\cdot 3 =$$

$$= -2 + \cancel{15} + 72 - 8 + 18 - \cancel{15} =$$

$$= 80$$

$$\Delta z = \begin{vmatrix} 1 & 1 & 6 \\ 3 & -1 & 2 \\ 4 & 3 & 5 \end{vmatrix} = 1(-1)5 + 6\cdot 3\cdot 3 + 4\cdot 1\cdot 2 -$$

$$- 6(-1)4 - 5\cdot 3\cdot 1 - 1\cdot 2\cdot 3 =$$

$$= -5 + 54 + 8 + 24 - 15 - 6 = 60$$

$$X = \frac{\Delta x}{\Delta} = \frac{-20}{20} = -1$$

$$y = \frac{\Delta y}{\Delta} = \frac{80}{20} = 4$$

$$z = \frac{\Delta z}{\Delta} = \frac{60}{20} = 3$$

Ответ: $(-1; 4; 3)$

Проверка:

$$\begin{cases} x + y + z = 6 \\ 3x - y + 3z = 2 \\ 4x + 3y - 2z = 5 \end{cases}$$

$$\begin{cases} -1 + 4 + 3 = 6 \\ 3(-1) - 4 + 3 \cdot 3 = 2 \\ 4(-1) + 3 \cdot 4 - 3 = 5 \end{cases}$$

3) $\begin{cases} x+y-3z=-3 \\ 2x-y-6z=9 \\ -x+y+4z=-6 \end{cases}$

$$\Delta = \begin{vmatrix} 1 & 1 & -3 \\ 2 & -1 & -6 \\ -1 & 1 & 4 \end{vmatrix} = 1(-1)4 + (-3)2\cdot 1 + (-1)1(-6) -$$

$$-(-3)(-1)(-1) - 4\cdot 2\cdot 1 - 1(-6)1 =$$

$$= -4 - \cancel{6} + \cancel{6} + 3 - 8 + 6 =$$

$$= -3$$

$\Delta \neq 0 \Rightarrow$ единственное решение

$$\Delta x = \begin{vmatrix} -3 & 1 & -3 \\ 9 & -1 & -6 \\ -6 & 1 & 4 \end{vmatrix} = (-3)(-1)4 + (-3)9\cdot 1 + (-6)1(-6) -$$

$$- (-3)(-1)(-6) - 4\cdot 9\cdot 1 - (-3)1(-6) =$$

$$= 12 - 27 + \cancel{36} + \cancel{18} - \cancel{36} - \cancel{18} =$$

$$= -15$$

$$\Delta y = \begin{vmatrix} 1 & -3 & -3 \\ 2 & 9 & -6 \\ -1 & -6 & 4 \end{vmatrix} = 1\cdot 9\cdot 4 + (-1)(-3)(-6) + (-3)2(-6) -$$

$$- (-3)9(-1) - 4(-3)2 - 1(-6)(-6) =$$

$$= 36 - 18 + \cancel{36} - 27 + 24 - \cancel{36} =$$

$$= 15$$

$$\Delta z = \begin{vmatrix} 1 & 1 & -3 \\ 2 & -1 & 9 \\ -1 & 1 & -6 \end{vmatrix} = 1(-1)(-6) + (-3)2\cdot 1 + (-1)1\cdot 9 -$$

$$- (-3)(-1)(-1) - (-6)2\cdot 1 - 1\cdot 1\cdot 9 =$$

$$= \cancel{6} - \cancel{6} - 9 + 3 + 12 - 9 =$$

$$= 15 - 18 = -3$$

$$X = \frac{\Delta x}{\Delta} = \frac{-15}{-3} = 5$$

Ответ: $(5; -5; 1)$

$$y = \frac{\Delta y}{\Delta} = \frac{15}{-3} = -5$$

$$z = \frac{\Delta z}{\Delta} = \frac{-3}{-3} = 1$$

Проверка:

$$\begin{cases} x + y - 3z = -3 \\ 2x - y - 6z = 9 \\ -x + y + 4z = -6 \end{cases}$$

$$\begin{cases} 5 - 5 - 3(1) = -3 \\ 2 \cdot 5 + (-5) - 6(1) = 9 \\ -5 - 5 + 4 \cdot 1 = -6 \end{cases}$$